

Li-Fi Based Automatic Traffic Signal Control for Emergency Vehicles

CHAPTER

Abstract—

India is one of the fastest developing country in the world. One of the disadvantages of this unprecedented growth is the ineffective traffic management. Light Fidelity (Li-Fi) is a form of bi-directional Visible Light Communication (VLC) in which light is modulated at speeds greater than a human eye can follow. Li-Fi can be used to implement Vehicle to Vehicle (V2V) communication as it has many advantages over other communication protocols. One of the main advantages of Li-Fi is that it provides connectivity within a very large area with more security and higher data rates. Emergency vehicles such as Ambulances, Firefighting trucks, Police vehicles etc. can propagate faster through traffic-dense roads using Li-Fi based V2V communication system. In this paper, a V2V communication system or V2 traffic light communication is proposed in which the headlight and tail light of a vehicle are made as Li-Fi transmitter and Li-Fi receiver respectively. Using the proposed system, when a traffic signal receives an alert regarding an emergency vehicle, the traffic signal would immediately turn green. Thus, decreasing the waiting time of emergency vehicles in traffic dense lanes. The proposed system was designed, implemented and tested for its operation.

CHAPTER INTRODUCTION

Wireless communication has been solely based on radio waves. Due to technological developments, this spectrum has become congested. Similarly, the traffic density has also increased many folds. To have an efficient traffic management, we need reliable wireless V2V communication system. The choice of radio waves for this application involves a huge bandwidth to be allocated to implement V2V communication system, which is very difficult as this spectrum is already congested due to human cellular communication systems [1]. Also, if we consider the circuitry associated with this type of communication, it is expensive. Therefore, the need for an alternative spectrum is increasing day by day. Gamma rays, X-rays, Infra-red and Ultraviolet rays cannot be used to implement V2V communication system due to their harmful effects on human body.

VLC is an emerging communication platform which uses visible light in the frequency.

Li-Fi is a VLC protocol, which is like the Wi-Fi protocol in every layer except the physical layer [2]. The concept of Li-Fi was introduced by Dr. Harald Haas during his speech at TED Global 2011[3]. He considered LED bulbs as the principle source of light because older sources such as incandescent and fluorescent light bulbs have proved themselves to be inefficient in terms of luminosity and energy consumption when compared to LEDs (or Laser LED) . Their intensity can be modulated at very high speeds. This property is exploited in Li-Fi technology. Li-Fi also enables transmission of thousands of data streams in parallel at high speeds. A photodiode is used as the receiving element in this technology. Both these devices are cheap to acquire, which makes Li-Fi a economically viable solution to all the problems offered by radio wave communication.

CHAPTER LITERATURE SURVEY

Research on traffic management, V2V communication, Ambulance based traffic control has been carried out since 1990s. Various communication protocols like GSM, RFID, IR, ZigBee etc. have been considered to implement emergency vehicle based traffic control. However, the above methods cannot be used in today's world for various reasons.

GSM based communication a system has been extensively used for cellular communication. The infrastructure needed for GSM communication is high compared to other communication protocols.

Even though basic infrastructure for GSM is available, the use of this technology for V2V communication requires high bandwidth. RFID based V2V communication do not require line of sight. RFID based traffic control can detect emergency vehicles if it is in the frequency range of the traffic control. The prime disadvantage of RFID based V2V communication is low security offered where RFID messages are prone to unauthorized access. The metallic chassis of the vehicle can also cause problems to the RFID transceiver. IR based emergency vehicle detection is Unreliable as IR signals require line of sight and sunlight can cause interference with IR signals. ZigBee based traffic management is a good alternative when compared to other communication protocols mentioned above. It offers wide range of communication does not

require line of sight and is comparatively economical. However, to implement V2V communication or traffic management using ZigBee involves restrictions on the number of nodes per controller. Various research papers have proposed the use of Li-Fi for emergency vehicle based traffic control systems. It does not need any infrastructure like GSM. It offers a much secure communication compared to RFID. It is relatively economical compared to ZigBee. It does not use the congested 2.4 GHz frequency like ZigBee. In this paper, Li-Fi based V2V communication system is implemented for smart control of traffic signal.

The idea of Li-Fi was introduced by Harald Haas, the teacher of University of Edinburgh in Scotland and he brought the idea of "Wireless data from every light". Haas and his re-search assistant, Mostafa Afgani, first sent data by using light signals in 2007 [1]. The term Li-Fi was first used by Haas in his TED Global talk on Visible Light Communication. He referred this light signal as D-Light which can be used to produce data rates higher than 10 megabits per second which is much faster than our average broadband connection [2].

Various research papers have proposed the use of Li-Fi for emergency vehicle based traffic control systems [5,6,7]. It does not need any infrastructure like GSM. It offers a much secure communication compared to RFID. It is relatively economical compared to ZigBee. It does not use the congested 2.4 GHz frequency like ZigBee. In this paper, Li-Fi based V2V communication system is implemented for smart control of traffic signal.

Abhishek Patni, Bhavini Mishra, Harsh Aditya, Yogesh Kumar and Rohit Sharma presented light is everywhere street lamp, traffic light and meeting hall. Imagine that all these light bulbs are high speed wireless transmitter that connects either human to human or systems with systems. Light fidelity or we can say LIFI, a new WIFI is based on the very recent intelligence of light emitting diode (LED) technology as the source of lightning. LIFI is a fully networked, bidirectional high speed wireless connection system. Unlike wireless networks.

LIFI networks do not rely on any fixed frequency spectrum which is very limited. Instead, LIFI is based on visible light spectrum which provides the spectrum which is more than 10,000 times to radio wave to deliver desired data. These unique characteristics of LIFI pose a number of challenges for the implementation of capacity, efficiency, availability and security in the wireless network system. In this Topic, we study the LIFI use in highway navigation system thus understanding the vulnerabilities and how LIFI will use full in future. Further, we discuss the various issues in WIFI and explore approaches to solve the problems in communication.

V. K. G. Kalaiselvi, A. Sangavi and Dhivya presented wireless communication has become a basic utility in personal and business life such that it becomes a fundamental of our lives and this communication uses the radio spectrum for data transfer. There are issues in using the radio spectrum further, they are, capacity, efficiency, availability and security. The defects of Wi-Fi technology has given birth to the concept of Li-Fi (Light Fidelity) technology. Li-Fi can be defined as a light based Wi-Fi. This technology mainly serves the purpose of transmitting data using retrofitting of LED Bulbs that has high efficiency, durability and reliability. This technology emerged by 1990s in a lot of countries like Japan, German and of vehicles with the traffic light system in order to prioritize the vehicles and change the signals accordingly rather than by a process of pre-defined order or by manual order. Traffic lights already use LED lighting, so that this proposed system may seem easy to implement. Sending data through siren lights in an ambulance and fire extinguishers to a traffic light control system and switching the signal in order to allow faster and non-interrupted transport. Here is already a kind of system in play for GPS navigational systems, but the traffic lights can be used in updating drivers using basic information or streaming video directly from news broadcasts i.e, about the accidents or delays up ahead. Research on traffic management, V2V communication, Ambulance based traffic control have been carried out since 1990s. Various communication protocols like GSM, RFID, IR, ZigBee etc. have been considered to implement emergency vehicle based traffic control. However, the above methods cannot be used in today's world for various reasons.

J. Saradha, G. Vijayshri and T. Subha presented road traffic congestion becomes a major issue for highly crowded metropolitan cities like, Chennai. Ambulance service is one of the major services which gets affected by traffic jams. To smoothen the ambulance movement this Topic have come up with the solution of "Intelligent automatic traffic control for ambulance". The proposed system creates a android app that connects both the ambulance and the traffic signal station using cloud network.

This system makes uses RFID (radio frequency identification) technology to implement the Intelligent traffic signal control. The basic idea behind the proposed system is, if the Ambulance halts on the way due to a traffic signal, RFID installed at the traffic signal tracks the RFID

tagged ambulance and sends the data to the cloud. After the acknowledgment for the user through the mobile app, the particular signal is made Green for some time and after the ambulance passes by, it regains its original flow of sequence of signaling. If, this scheme is fully automated, it finds the ambulance spot, controls the traffic lights. This system controls the traffic lights and saves the time in emergency periods. Thus it acts as a life saver project.

CHAPTER

PROBLEM STATEMENT

India is one of the fastest developing countries in the world. One of the disadvantages of this unprecedented growth is the ineffective traffic management. The problem of lack of radio spectrum, space and low internet connection speed.

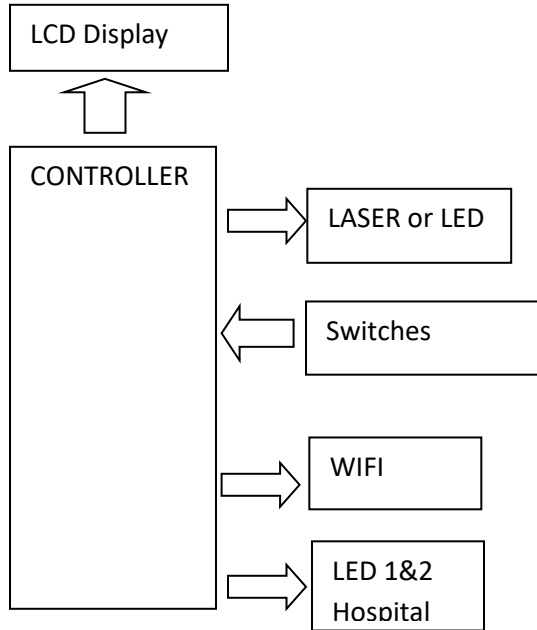
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OBJECTIVES OF THE PROJECT

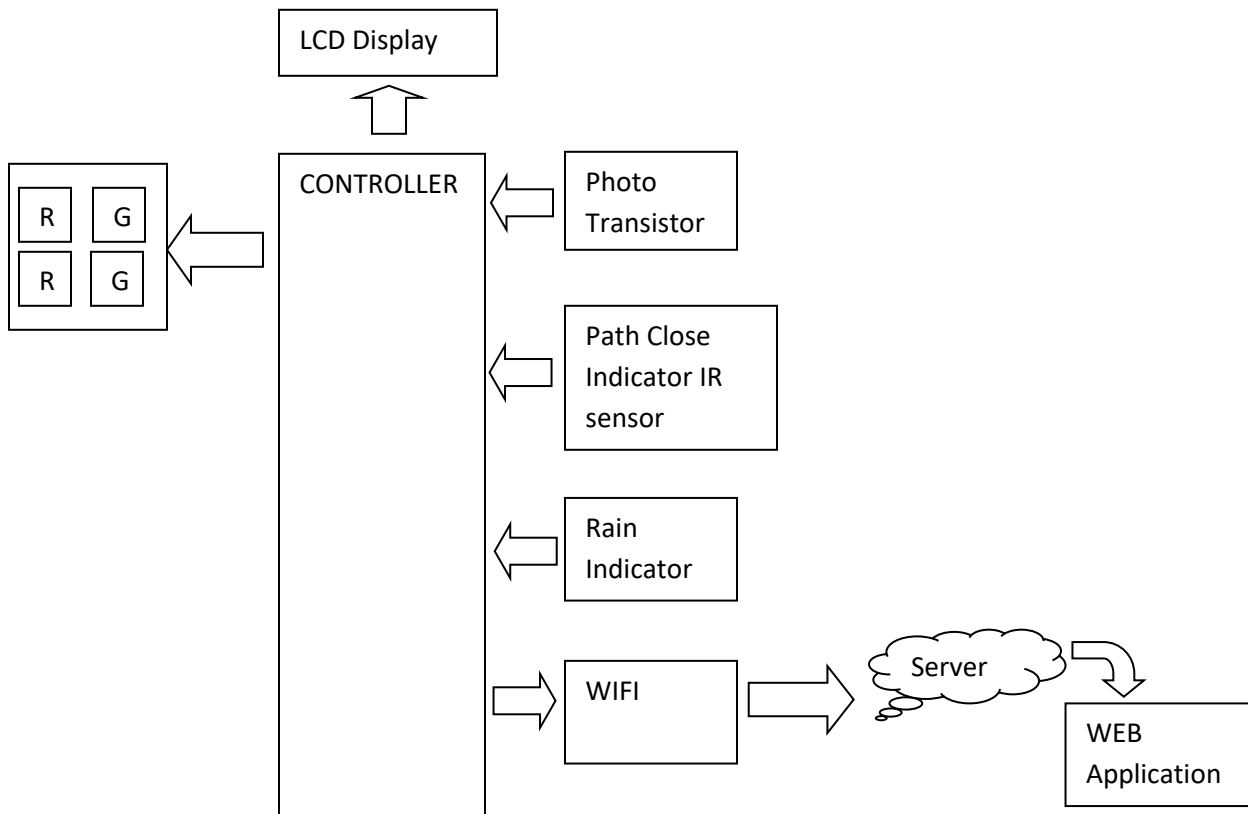
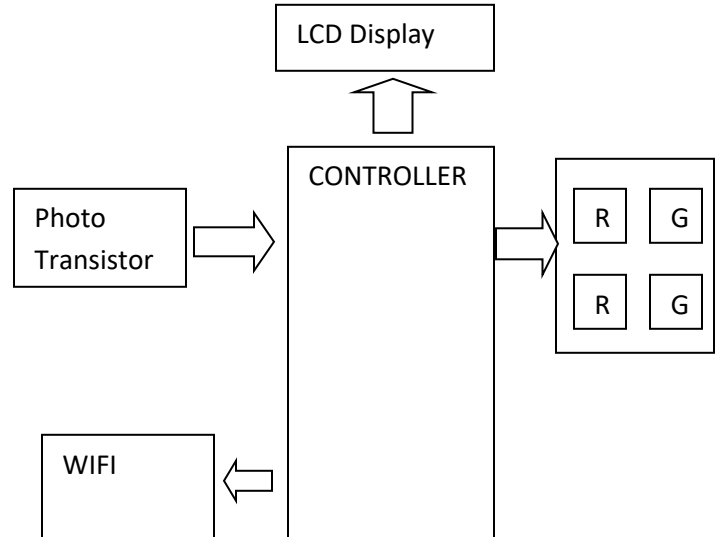
1. The main objective of our project is to provide the traffic clearance for emergency vehicles.
2. Get updates traffic from environment parameters
3. To provide maximum communication for ambulance from environment and hospitals.
4. To provide immediate response from the hospital side for emergency vehicle.

CHAPTER BLOCK DIAGRAM:

Ambulance:



Traffic1:



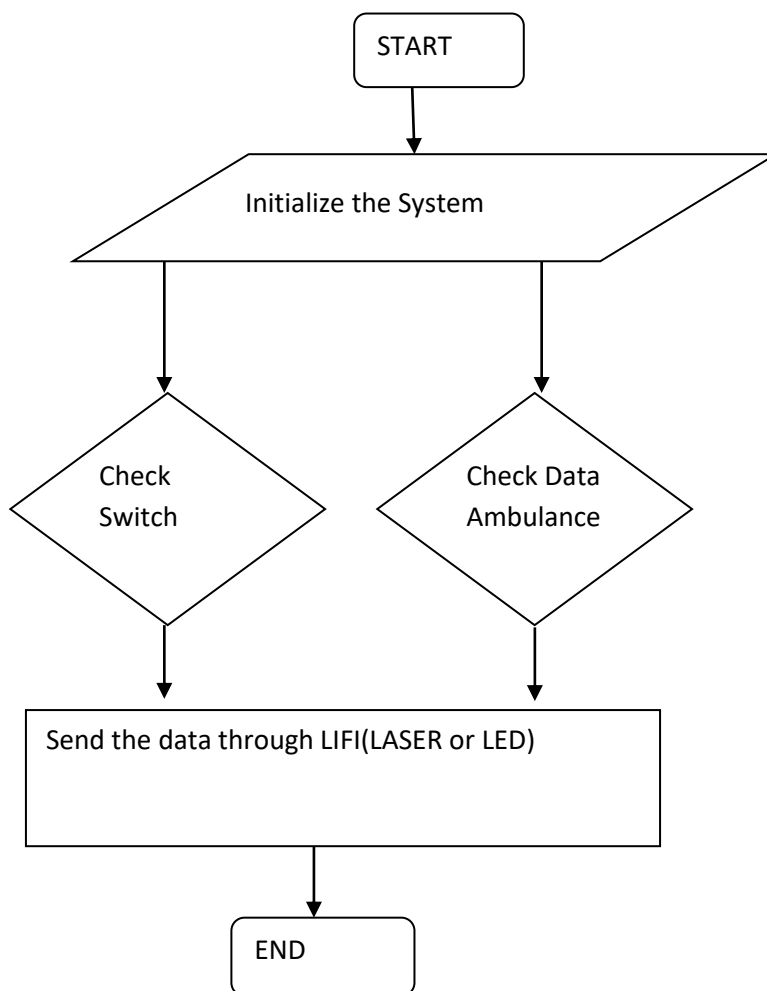
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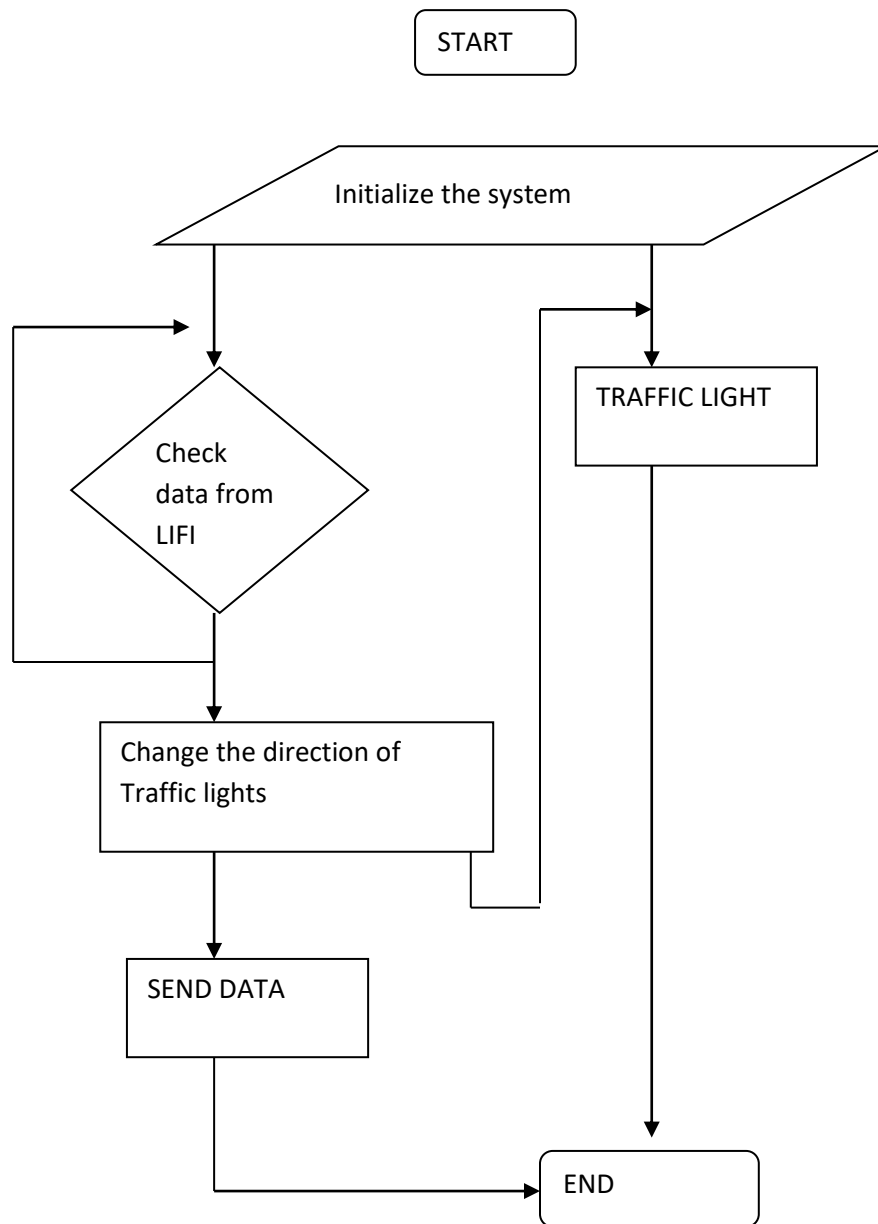
EXPECTED RESULTS

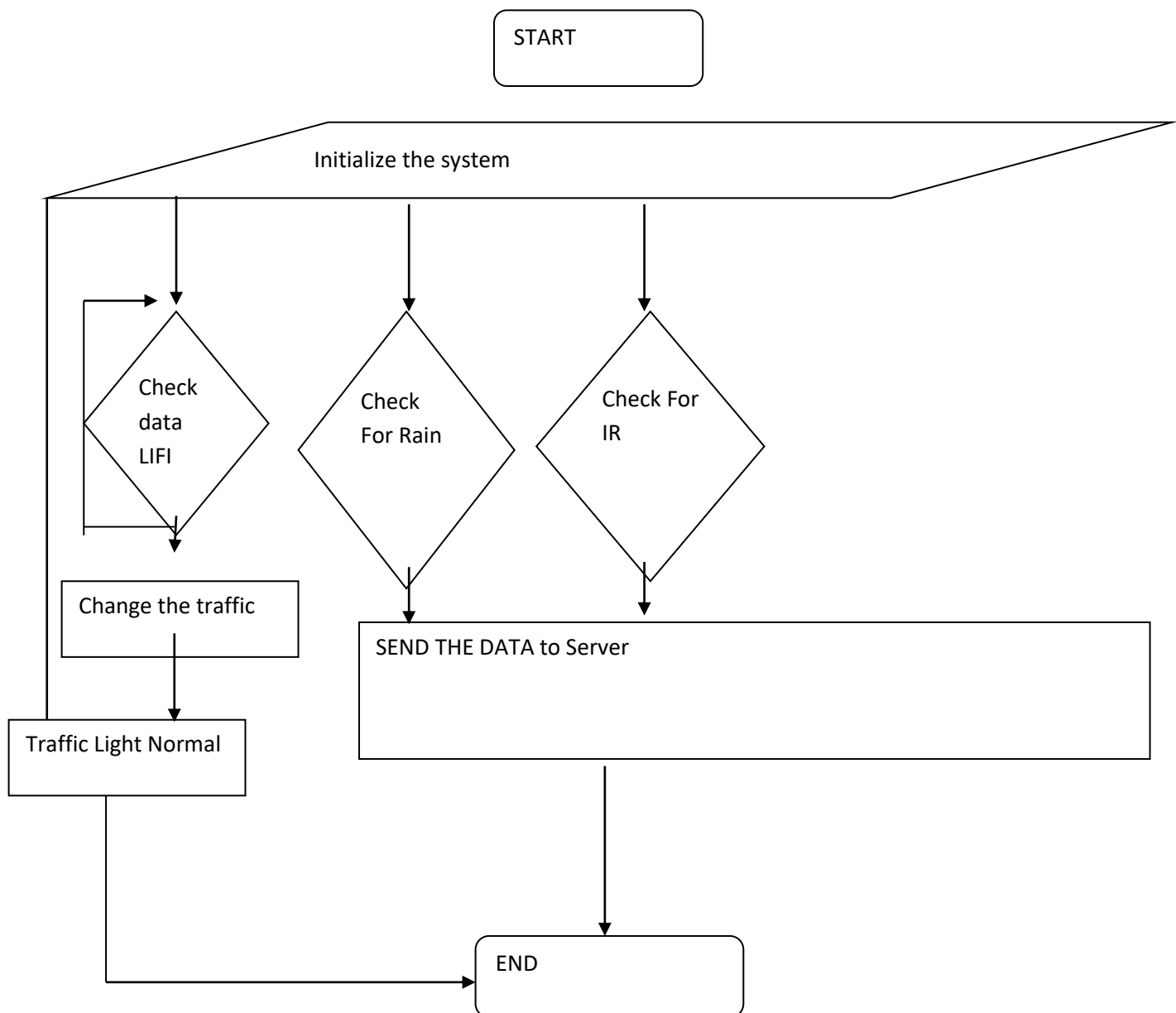
In this Topic, a single hop scenario need to be implemented. The time taken for the alert message to propagate from an emergency vehicle to a non-emergency vehicle and then to the traffic signal control is approximately one second. This time can be further reduced by implementing the same functionality using optimized code. The time can also be reduced by designing better modulation and demodulation circuits which supports higher data rates. The alert message can also be encrypted to make it secure.

CHAPTER

FLOW CHART:







HARDWARE and SOFTWARE:

HRADWARE:

1.Controller(8051 or Arduino Uno)

2.LASER OR LED

3.UDN

4.Phototransistor

5.Rain sensor

6. IR SENSOR

7.

8.LCD DISPLAY

9.WIFI

SOFTWARE:

1.Keil

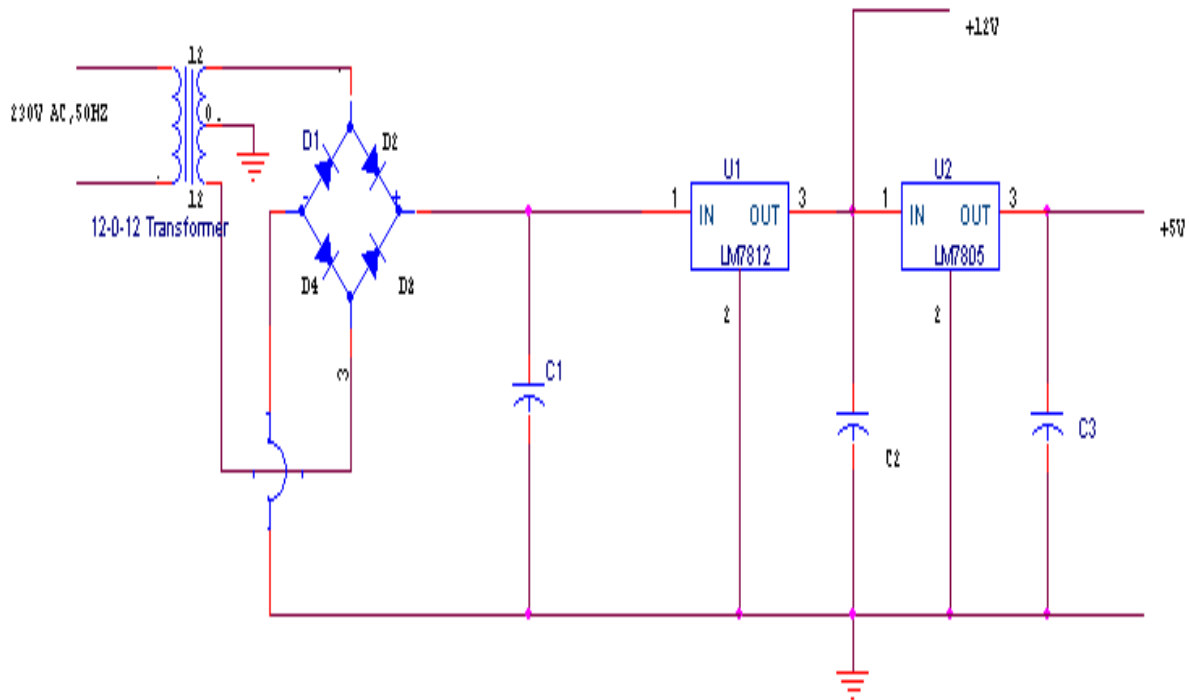
2.ARDUINO IDE

3.Flash MAGIC

4.ADAFRUIT IO(WEB application)

HARDWARE:

POWER SUPPLY UNIT



In every project we need different voltages for different circuits. So we need to construct different power supply of different voltages employing different voltage transformers, rectifier circuits, filter circuits and regulator circuits.

This type of construction requires many components (transformers, capacitors, regulators.....etc.). So the size of the power supply becomes bulky and costly. To overcome above disadvantages by using regulator IC'S the different voltages (12V, 9V.....etc.) can be obtained with only one transformer.

The circuit diagram of Dual power supply is shown in the figure 3.1. The function of each component of the circuit is explained below. The circuit consists of following stages.

1. Transformer
2. Rectifier
3. Filter
4. Regulator

TRANSFORMER:

It is an electrical device which transfers the power from one winding to the other winding with isolation. All the electronic gadgets work for less voltage (normally 3V to 12V). So a step

down transformer is used, whose function is to step down the AC voltage from 230V to required voltage depending on the need. In this project 12V-0-12V is used. The output of transformer is 12V AC which is connected to the diodes for rectification.

RECTIFIER CIRCUIT:

It employs diodes, which converts AC voltage into DC voltage. The output of rectifier circuit is not a pure DC. It also consists of some AC components, which is called ripples. In order to remove these AC components, filter circuits are employed. So the output of rectifier circuit is fed to the filter circuit (capacitor).

FILTER CIRCUIT:

Filter circuit employs electrolytic capacitors in order to remove the AC components. As we know the capacitor does not allow DC components to pass through it because it offers high reactance to the DC component. And offers less reactance to the AC component, so all AC components will be bypasses through the capacitors to ground.

REGULATOR:

Regulator is an electronic circuit whose function is to keep output always constant though the input is varied. In this project the three terminal IC regulators of 7812 & 7805 is used for providing output DC voltages. E.g. 7809, the number 78 represents the positive regulator IC and 09 represents the output voltage i.e. output is 12V.

Component List:

SL.NO.	Component Name	Specification	Value	Quantity
1.	Transformer	Step-down	12-0-12	1
2.	Diode	D1	IN4001	1
		D2	IN4001	1

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		D3	IN4001	1
		D4	IN4001	1
3.	Capacitors	C1	1000uF,25V	1
		C2	0.01uF	1
		C3	0.01uF	1
4.	Regulator	U1	LM7812	1
		U2	LM7805	1

Arduino Uno



Overview

The Arduino Uno is a microcontroller board based on the ATmega328 (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It

contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.

The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter. Revision 2 of the Uno board has a resistor pulling the 8U2 HWB line to ground, making it easier to put into DFU mode. Revision 3 of the board has the following new features:

- 1.0 pinout: added SDA and SCL pins that are near to the AREF pin and two other new pins placed near to the RESET pin, the IOREF that allow the shields to adapt to the voltage provided from the board. In future, shields will be compatible both with the board that use the AVR, which operate with 5V and with the Arduino Due that operate with 3.3V. The second one is a not connected pin, that is reserved for future purposes.
- Stronger RESET circuit.
- Atmega 16U2 replace the 8U2.

"Uno" means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduino boards, and the reference model for the Arduino platform; for a comparison with previous versions, see the index of Arduino boards.

Summary	ATmega328
Microcontroller	
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limits)	6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	6
DC Current per I/O Pin	40 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB (ATmega328) of which 0.5 KB used

	by bootloader
SRAM	2 KB (ATmega328)
EEPROM	1 KB (ATmega328)

Clock Speed 16 MHz

Power

The Arduino Uno can be powered via the USB connection or with an external power supply. The power source is selected automatically.

External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery. The adapter can be connected by plugging a 2.1mm center-positive plug into the board's power jack. Leads from a battery can be inserted in the Gnd and Vin pin headers of the POWER connector.

The board can operate on an external supply of 6 to 20 volts. If supplied with less than 7V, however, the 5V pin may supply less than five volts and the board may be unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

The power pins are as follows:

- **VIN.** The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.
- **5V.** This pin outputs a regulated 5V from the regulator on the board. The board can be supplied with power either from the DC power jack (7 - 12V), the USB connector (5V), or the VIN pin of the board (7-12V). Supplying voltage via the 5V or 3.3V pins bypasses the regulator, and can damage your board. We don't advise it.
- **3V3.** A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- **GND.** Ground pins.

Memory

The ATmega328 has 32 KB (with 0.5 KB used for the bootloader). It also has 2 KB of SRAM and 1 KB of EEPROM (which can be read and written with the EEPROM library).

Input and Output

Each of the 14 digital pins on the Uno can be used as an input or output, using pinMode(), digitalWrite(), and digitalRead() functions. They operate at 5 volts. Each pin can provide or receive a maximum of 40 mA and has an internal pull-up resistor (disconnected by default) of 20-50 kOhms. In addition, some pins have specialized functions:

- **Serial: 0 (RX) and 1 (TX).** Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip.
- **External Interrupts: 2 and 3.** These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the `attachInterrupt()` function for details.
- **PWM: 3, 5, 6, 9, 10, and 11.** Provide 8-bit PWM output with the `analogWrite()` function.
- **SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK).** These pins support SPI communication using the SPI library.
- **LED: 13.** There is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

The Uno has 6 analog inputs, labeled A0 through A5, each of which provide 10 bits of resolution (i.e. 1024 different values). By default they measure from ground to 5 volts, though it is possible to change the upper end of their range using the AREF pin and the `analogReference()` function. Additionally, some pins have specialized functionality

- **TWI: A4 or SDA pin and A5 or SCL pin.** Support TWI communication using the Wire library.
- There are a couple of other pins on the board:
- **AREF.** Reference voltage for the analog inputs. Used with `analogReference()`.
- **Reset.** Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

Communication

The Arduino Uno has a number of facilities for communicating with a computer, another Arduino, or other microcontrollers. The ATmega328 provides UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The '16U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on Windows, a .inf file is required. The Arduino software includes a serial monitor which allows simple textual data to be sent to and from the Arduino board. The RX and TX LEDs on the board will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer (but not for serial communication on pins 0 and 1).

A SoftwareSerial library allows for serial communication on any of the Uno's digital pins.

The ATmega328 also supports I2C (TWI) and SPI communication. The Arduino software includes a Wire library to simplify use of the I2C bus; see the documentation for details. For SPI communication, use the SPI library.

Programming

The Arduino Uno can be programmed with the Arduino software (download). Select "Arduino Uno" from the **Tools > Board** menu (according to the microcontroller on your board). For details, see the reference and tutorials.

The ATmega328 on the Arduino Uno comes preburned with a bootloader that allows you to upload new code to it without the use of an external hardware programmer. It communicates using the original STK500 protocol (reference, C header files).

You can also bypass the bootloader and program the microcontroller through the ICSP (In-Circuit Serial Programming) header; see these instructions for details.

The ATmega16U2 (or 8U2 in the rev1 and rev2 boards) firmware source code is available . The ATmega16U2/8U2 is loaded with a DFU bootloader, which can be activated by:

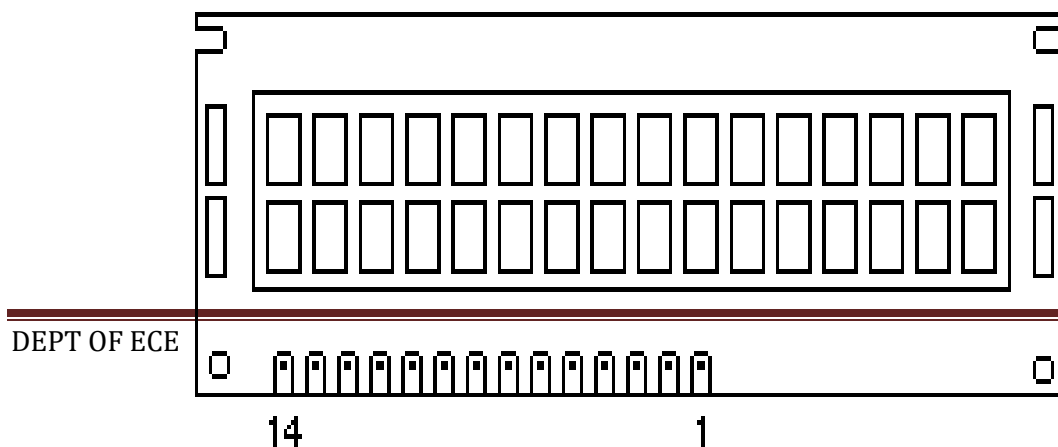
- On Rev1 boards: connecting the solder jumper on the back of the board (near the map of Italy) and then resetting the 8U2.
- On Rev2 or later boards: there is a resistor that pulling the 8U2/16U2 HWB line to ground, making it easier to put into DFU mode.

You can then use Atmel's FLIP software (Windows) or the DFU programmer (Mac OS X and Linux) to load a new firmware. Or you can use the ISP header with an external programmer (overwriting the DFU bootloader). See this user-contributed tutorial for more information.

Physical Characteristics

The maximum length and width of the Uno PCB are 2.7 and 2.1 inches respectively, with the USB connector and power jack extending beyond the former dimension. Four screw holes allow the board to be attached to a surface or case. Note that the distance between digital pins 7 and 8 is 160 mil (0.16"), not an even multiple of the 100 mil spacing of the other pins

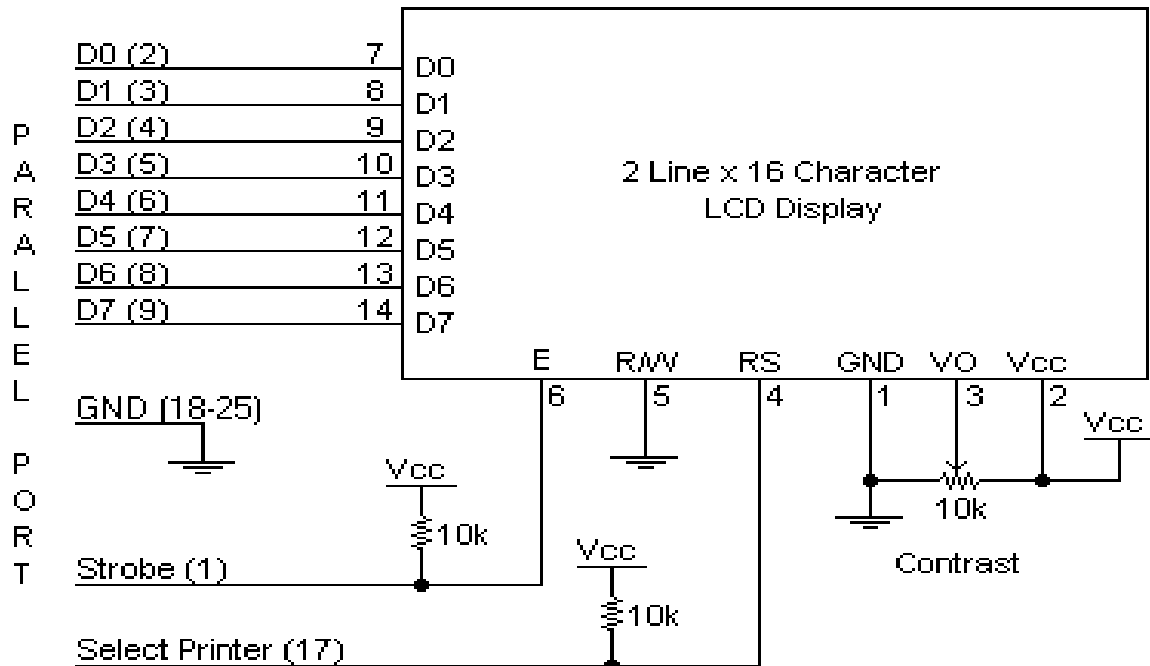
LCD OPERATION:



In recent years the LCD is finding widespread use replacing LEDs (seven segment LEDs or other multisegment LEDs). This is due to the following reasons:

1. The declining prices of LDCs.
2. The ability to display numbers, characters, and graphics. This is in contrast to LEDs, Which are limited to numbers and a few characters.
3. Incorporation of a refreshing controller into the LDC, thereby relieving the CPU of the task of refreshing the LCD. In contrast, the LED must be refreshed by the CPU (or in some other way) to keep displaying the data.
4. Ease of programming for characters and graphics.

6.2 LCD pin descriptions:



Pin Description for LCD

The LCD discussed in this section has 14 pins. The function of each pin is given in Table 11.1. Figure 11.1 shows the pin positions for various sample LCDs.

VCC, VSS, and VEE:

While V_{CC} and V_{SS} provide +5V and ground, respectively, V_{EE} is used for controlling LCD contrast. RS, register select. There are two very important registers

Inside the LCD. The RS pin is used for their selection as follows. If RS=0, the instruction command code register is selected, allowing the user to send a command such as clear display, cursor at home, etc. IF RS=1 the data register is selected, allowing the user to send data to be displayed on the LCD.

R/W (read/write):

R/W input allows the user to write information to the LCD or read information from it. R/W=1 when reading; R/W=0 when writing.

E(enable):

The enable pin is used by the LCD to latch information presented to its data pins. When data is supplied to data pins, a high-to-low pulse must be applied to this pin in order for the LCD to latch in the data present at the data pins. This pulse must be a minimum of 450 ns wide.

D0 - D7:

The 8-bit data pins, D0 - D7, are used to send information to the LCD or read the contents of the LCD's internal registers.

To display letters and numbers, we send ASCII codes for the letters A - Z, a - z, and numbers 0 - 9 to these pins while making RS=1.

There are also instruction command codes that can be sent to the LCD to clear the display or force the cursor to the home position or blink the cursor. Table 11.2 lists the instruction command codes.

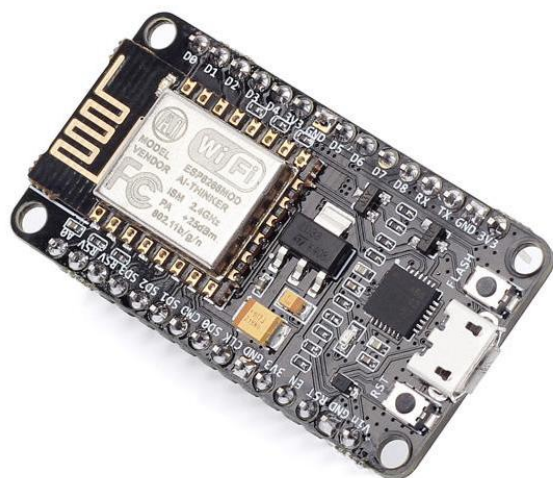
We also use RS = 0 to check the busy flag bit to see if the LCD is ready to receive information. The busy flag is D7 and can be read when R/W=1 and RS = 0, as follows: if R/W = 1, RS = 0. When D7 = 1 (busy flag = 1), the LCD is busy taking care of internal operations and will not accept any new information. When D7 = 0, the LCD is ready to receive new information. Note: It is recommended to check the busy flag before writing any data to the LCD.

Pin Descriptions for LCD:

Pin	Symbol	I/O	Description
1	VSS	--	Ground
2	VCC	--	+5V power supply
3	VEE	--	Power supply to control contrast

4	RS	I	--	RS=0 to select command register, RS=1 to select data register
5	R/W	I		R/w-0 for write, R/W=1 for read
6	E	I/O		Enable
7	DB0	I/O		The 8-bit data bus
8	DB1	I/O		The 8-bit data bus
9	DB2	I/O		The 8-bit data bus
10	DB3	I/O		The 8-bit data bus
11	DB4	I/O		The 8-bit data bus
12	DB5	I/O		The 8-bit data bus
13	DB6	I/O		The 8-bit data bus
14	DB7	I/O		The 8-bit data bus

NodeMCU v2 WIFI MODULE



The NodeMcu is an open-source firmware and development kit that helps you to Prototype your IOT product within a few Lua script lines.

Features:

- Open-source
- Interactive
- Programmable
- Low cost
- Simple
- Smart
- WI-FI enabled

Arduino-like hardware IO

Advanced API for hardware IO, which can dramatically reduce the redundant work for configuring and manipulating hardware.

Code like arduino, but interactively in Lua script.

Nodejs style network API

Event-driven API for network applications, which facilitates developers writing code running on a 5mm*5mm sized MCU in Nodejs style.

Greatly speed up your IOT application developing process.

Specification:

The Development Kit based on ESP8266, integrates GPIO, PWM, IIC, 1-Wire and ADC all in one board.

Power your development in the fastest way combining with NodeMCU Firmware!

- USB-TTL included, plug&play
- 10 GPIO, every GPIO can be PWM, I2C, 1-wire
- FCC CERTIFIED WI-FI module (Coming soon)

PCB antenna

LI - FI MODULE

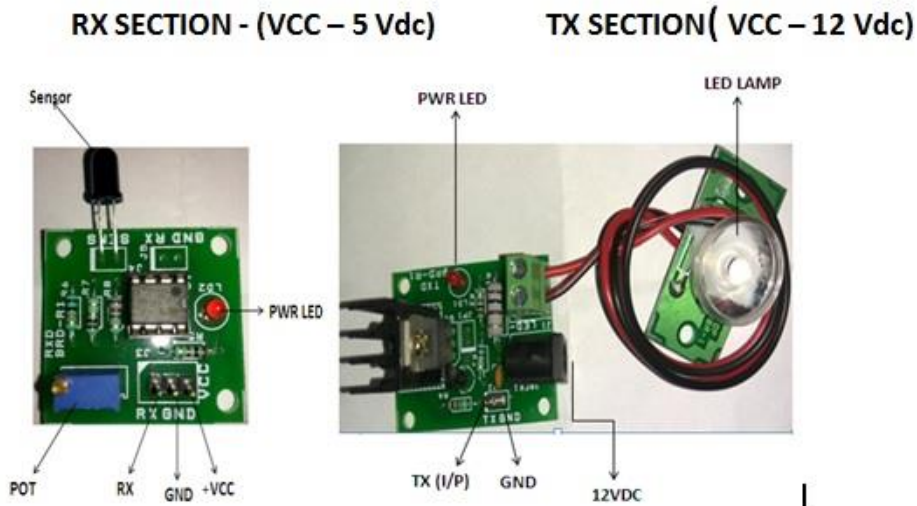
Description :

Li-Fi stands for Light Fidelity .Li-Fi technology provides transmission of data through illumination by sending data through LED that varies in intensity faster than the human eye can follow. This product focus on developing a Li-Fi based system and analyze its performance with respect to existing technology. The heart of this technology is a new generation of high brightness LEDs. The product consisting of a transmitter which includes a light source and the receiver circuit which receives the data transmitted via light waves. The same system can be employed in industries making industrial automation using

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the existing light source a reality. A biomedical sensor is also introduced herewith which helps in diagnosis of a patient's medical conditions and is quickly made available to all the emergency points through light.

LIFI CONNECTION DETAILS



Testing Procedure

- Connect Power Supply 5vdc to reciver / 12 VDC to Transmitter
- Supports Baudrate – 1200 & 2400
- Keep Straight as TX LED to RX Reciver (Max Distance 20 Cm)

Rain



- Comparator output signal clean wave good driving ability, than 15mA;
- Power adjust sensitivity; Rated voltage and 3.3V-5V
- Small Board PCB dimensions: 3.2cm x 1.4cm, Power indicator light, the output signal LED indicating lamp.
- Using wide LM393 voltage comparator
- Output: digital switching output (0 and 1) and AO analog voltage output; With fixing bolt holes for easy installation

Specifications for this item

Brand Name	OLatus	
Colour	multicolour	
Item Weight	15.0 grams	
Manufacturer Number	Series	OL-SENSOR-RAIN

Model Number	OL-SENSOR-RAIN
Number of Items	1

MQ 5 SENSOR



FEATURES

- * High sensitivity to Carbon,LPG, natural gas , town gas
- * Small sensitivity to alcohol, smoke.
- * Fast response . * Stable and long life * Simple drive circuit

APPLICATION

They are used in gas leakage detecting equipments in family and industry, are suitable for detecting of

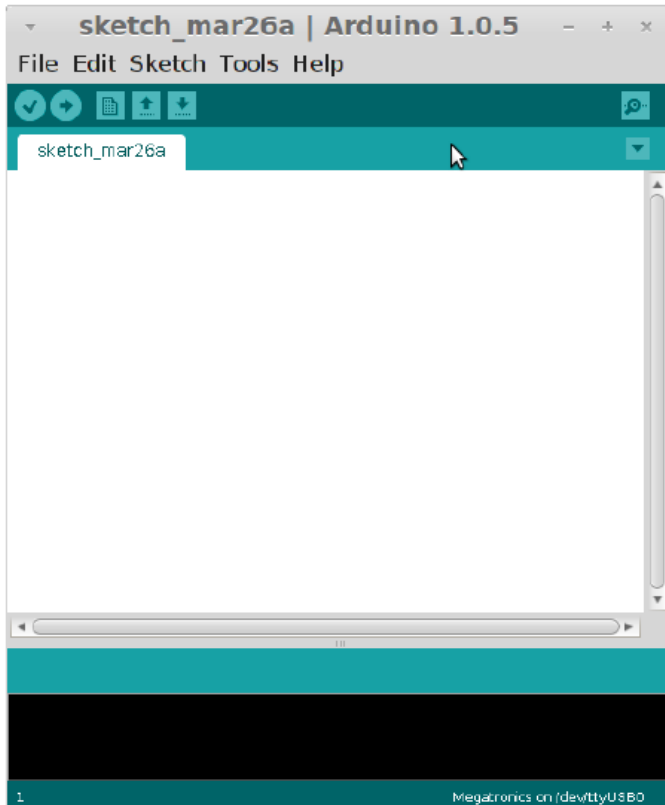
LPG, natural gas , town gas, avoid the noise of alcohol and cooking fumes and cigarette smoke.

QUICK OVERVIEW

1. Operating Voltage: 5V DC
2. High sensitivity to LPG, natural gas, town gas.
3. Small sensitivity to alcohol, smoke.
4. Stable and long life.
5. Simple drive circuit.
6. Fast response.

SOFTWARE

Arduino IDE



Arduino is an open-source project, enabling hobbyists to easily take advantage of the powerful Atmega chips. The Arduino IDE is the software where you can write code and upload it to the Atmega chip. The code is then executed on the chip.

Most 3D-printer electronics are Arduino-compatible, they use the Atmega chip and enable the user to upload their code using Arduino. This includes Megatronics, Minitronics and RAMPS.

Before you can start using the electronics you need software 'firmware', that translates machine instructions (gcode) into actual movements. There are a few options here, including Marlin and Sprinter and Repetier. The actual firmware is not discussed in this document. You can use Arduino to upload this firmware onto your electronics. This document will guide you in the steps you need to take.

To upload a firmware, you must first open the files using File → Open. Select the .ino file from the directory containing the firmware. Arduino will open several tabs with files.

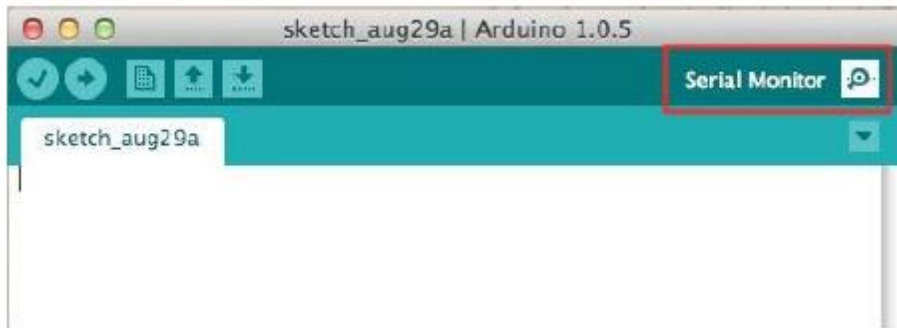
Next step is to select the correct electronics board. From the Tools menu, locate the Board item. This item should include a few sub items, including Megatronics, Minitronics, Arduino mega 2560 (RAMPS with mega 2560) and Arduino Mega 1280 (RAMPS with mega1280). Select the board that fits your electronics.

Also we need to select the serial port the electronics is connected to. In the Tools menu, locate the Serial port item. This should include at least one item if the board is connected and the drivers are installed properly. If there are multiple items here, you need to find out which is the correct one by unplugging the board and checking which port was removed.

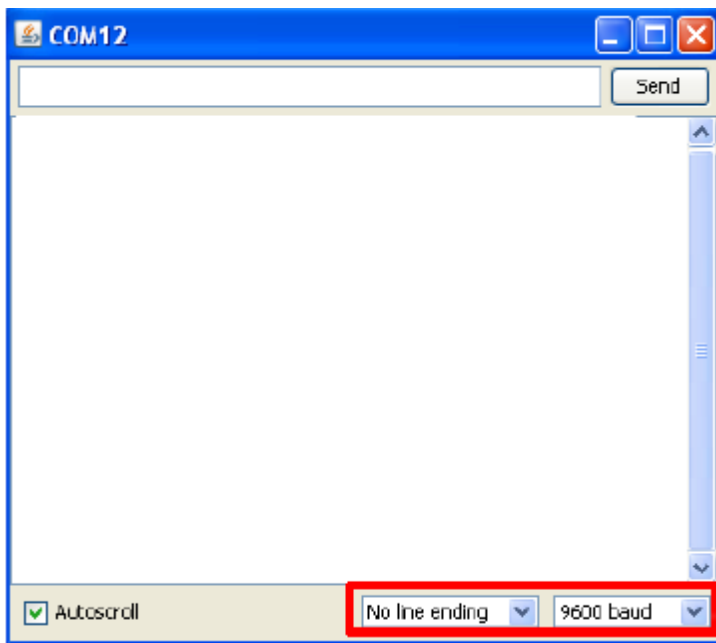
Li-Fi Based Automatic Traffic Signal Control for Emergency Vehicles

Once you have set the board and serial port, you can upload the firmware by pressing File → Upload. Arduino will try to compile the firmware, if any errors occur the process will stop and you will need to fix the errors before trying again. Once compilation is complete, the actual upload will start. This may take a minute for a large sketch. Arduino

You can actually 'talk' to the firmware using the Serial monitor. Make sure the correct serial port is selected and locate the Serial monitor button.



This will open a new window:



Make sure the line ending is set to newline and the baud rate corresponds to your firmware (115200 mostly), or you will get gibberish. You can enter a command in the upper box (M105 for example), this will result in response from the firmware. The temperature in this case

WEB APPLICATION

Adafruit IO

Overview

Adafruit IO is a system that makes data useful. Our focus is on ease of use, and allowing simple data connections with little programming required.

IO includes client libraries that wrap our REST and MQTT APIs. IO is built on Ruby on Rails, and Node.js. Adafruit IO is currently in beta. If you would like to join the beta, head over to io.adafruit.com to sign up (<http://adafru.it/eZ8>).

Please keep in mind that Adafruit IO is still in the beta testing stage. There will be bugs!

[Click Here to Submit a Bug or Feature Request](#)

<http://adafru.it/em2>

The client libraries are a work in progress. Please check back often for updates.

Getting Started

If you haven't already, log into your Adafruit account and then head over to io.adafruit.com (<http://adafru.it/fsU>) and click on the 'JOIN THE BETA LIST' button. We are slowly adding new beta users to help test Adafruit IO, and we will eventually open it up to everyone on the list.

Once you have been invited to the beta test, open up io.adafruit.com and you will be sent to a welcome dashboard.

Check out the following guides to understand the basics of creating a feed and a dashboard:

Adafruit IO Basics: Feeds (<http://adafru.it/ioA>)

Adafruit IO Basics: Dashboards (<http://adafru.it/f5m>)

Also check out the projects page (<http://adafru.it/iQB>) for a list of projects and examples to help understand the service.

Continue on reading this guide to learn about the client libraries that are available to send and receive data with Adafruit IO.

In addition you can learn about the protocols that Adafruit IO uses and how to use them with your own client code.

Client Libraries

The Adafruit IO client libraries greatly simplify interacting with the server. We have a few libraries built out already:

Arduino (<http://adafru.it/iQC>)

Step #0 - adafruit.io key and feeds

Before you can go crazy with Internetting your Things, you will need to do a little light config work to get adafruit.io ready for you.

To do this we'll introduce three new terms

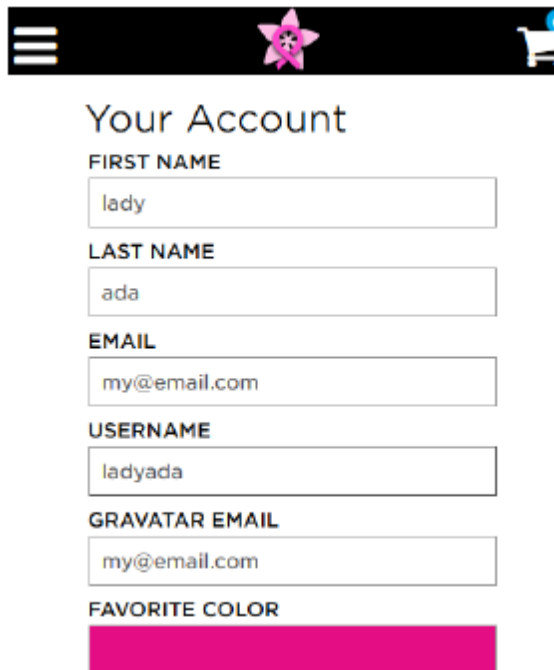
Account username - This is the name of your account, which you set when creating your adafruit account.

Key - this is a long, unique identifier that you use to authenticate any devices using your account. **This is your password! Keep it safe!** You get one key per account, but you can, at any time revoke and regenerate your key.

Feed - this is basically a set of data that you can read or write from like a sequential file. There is some history stored with feeds, with MQTT you cannot access historical data (REST does support it) but you can add data and you can receive the latest added data.

Where to find your username

You can find your username by visiting <https://accounts.adafruit.com/> (<http://adafru.it/dyy>) and logging in. Your username is right there!

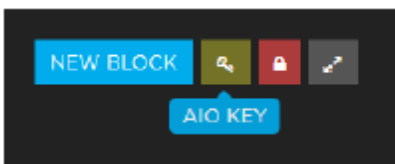


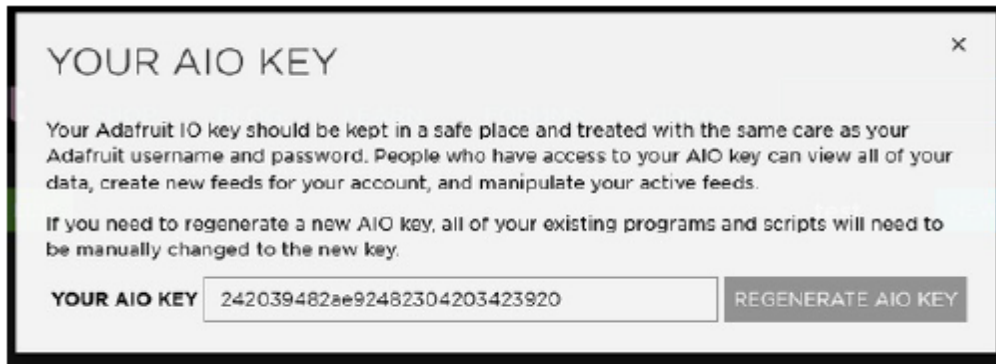
The screenshot shows the 'Your Account' form on the Adafruit website. At the top, there is a navigation bar with a hamburger menu icon, a pink star icon, and a shopping cart icon. The form itself is titled 'Your Account' and contains several input fields: 'FIRST NAME' with the value 'lady', 'LAST NAME' with the value 'ada', 'EMAIL' with the value 'my@email.com', 'USERNAME' with the value 'ladyada', and 'GRAVATAR EMAIL' with the value 'my@email.com'. Below these fields is a 'FAVORITE COLOR' section with a bright pink color bar selected.

Where to find your adafruit.io key

Visit <https://io.adafruit.com> (<http://adafru.it/tCt>) and look in the top left. You will see a little navbar with a yellow key

Click it to see your key. If you regenerate your key, your old key will no longer be valid and you'll need to update all your projects!





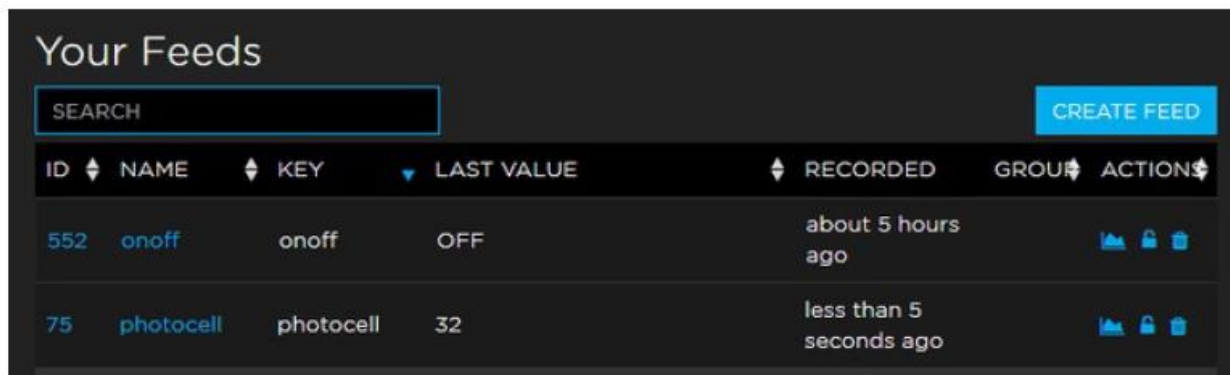
Create your first two feeds

You can read up on how to do this here

<https://learn.adafruit.com/adafruit-io-basics-feeds> (<http://adafru.it/ioA>)

Once you've read that. Go to your feeds page and create two feeds

1. **photocell** - this feed will store light data from your device to adafruit.io
2. **onoff** - this feed will act as an on/off switch, sending data to your device from adafruit.io



Create a dashboard

Like feeds, this has its very own, excellent tutorial. Read all about it here

[https://learn.adafruit.com/create-an-internet-of-things-dashboard-with-adafruit-dot-](https://learn.adafruit.com/create-an-internet-of-things-dashboard-with-adafruit-dot-io)

<http://adafru.it/iWe>

io

Once you've read it, create a dashboard with a **gauge** connected to **photocell**

CREATE A NEW BLOCK x

STEP 1: CHOOSE BLOCK TYPE EDIT

STEP 2: CHOOSE FEEDS

i Add up to 1 feed

NEW FEED NAME
CREATE

FEED/GROUP	LAST VALUE	RECORDED	ACTION
photocell	63	less than 5 seconds ...	CHOOSE

NEXT STEP >

Use a thin type gauge with min value 0 and max value 1024 (this could store a 10 bit value)

CREATE A NEW BLOCK x

STEP 1: CHOOSE BLOCK TYPE EDIT

STEP 2: CHOOSE FEEDS EDIT

STEP 3: BLOCK SETTINGS

BLOCK TITLE

GAUGE MIN VALUE

GAUGE MAX VALUE

GAUGE WIDTH

THIN
▼

GAUGE LABEL

BLOCK PREVIEW

photocell


CANCEL

CREATE BLOCK

Arduino

On an Arduino there are two different libraries you can use to access Adafruit IO. One library is based on the REST API, and the other library is based on the MQTT API. The difference between these libraries is that MQTT keeps a connection to the service open so it can quickly respond to feed changes. The REST API only connects to the service when a request is made so it's a more appropriate choice for projects that sleep for a period of time (to reduce power usage) and wake up only to send/receive data. If you aren't sure which library to use, try starting with the Adafruit MQTT library below.

Adafruit MQTT Client Library

To use Adafruit IO with the MQTT protocol on an Arduino you can use the Adafruit MQTT Arduino library (<http://adafru.it/fp6>).

This is a general-purpose MQTT library for Arduino that's built to use as few resources as possible so that it can work with platforms like the Arduino Uno. Unfortunately platforms like the Trinket 3.3V or 5V (based on the ATtiny85) have too little program memory to use the library-- stick with a Pro Trinket or better! The Adafruit MQTT library currently supports the following networking hardware/platforms:

Adafruit CC3000 (<http://adafru.it/iRa>)

Adafruit FONA (<http://adafru.it/dFz>)

ESP8266 Arduino (<http://adafru.it/eSH>)

Generic Arduino Client Interface (<http://adafru.it/fpb>) (including Ethernet shield and similar network hardware)

To install the library you can use the Arduino library manager (<http://adafru.it/flm>) or download the library from GitHub (<http://adafru.it/fp7>) and manually install it (<http://adafru.it/dNR>).

On some platforms the Adafruit MQTT library uses the hardware watchdog to help ensure sketches run reliably. You'll need to install the Adafruit SleepyDog sleep and watchdog library (<http://adafru.it/fp8>), again using either the Arduino library manager or with a manual install.

Finally make sure you have any required libraries for your network hardware installed, such as the CC3000 library (<http://adafru.it/cFn>) or FONA library (<http://adafru.it/dDC>).

Once the library is installed open or restart the Arduino IDE and check out the example code included with the library. These examples show the basic usage of the library, like how to connect to Adafruit IO, subscribe to receive changes to a feed, and how to send values to a feed.

PubSubClient MQTT Library

Another popular MQTT library for the Arduino is the PubSubClient MQTT library (<http://adafru.it/e1W>) and it works great to access Adafruit IO. Note that the library only works with networking libraries that support the Arduino Client interface. This means the library will work with the Ethernet shield, CC3000 or even ESP8266 Arduino, but not the FONA platform because it does not have a Client interface.

Below is a small example that shows using the PubSubClient library with the CC3000. To use this you will need the Adafruit CC3000 library (<http://adafru.it/cFn>) and PubSubClient library (<http://adafru.it/e1W>) installed in Arduino. Note that you'll need to change the following #define configuration lines at the top to fill in your wireless AP connection details and Adafruit IO username, key, and feed name:

```
#define WLAN_SSID "... your WiFi SSID..."
```

```
#define WLAN_PASS "... your WiFi password..."
```

```
#define ADAFRUIT_USERNAME "... your Adafruit username (see accounts.adafruit.com)..."
```

```
#define AIO_KEY "... your Adafruit IO key..."
```

```
#define FEED_PATH ADAFRUIT_USERNAME "/feeds/feed-name/"
```

The FEED_PATH is the path to publish or subscribe to for interacting with a feed. Notice that the path starts with the

Adafruit account name and is followed by "/feeds/feed-name" (where "feed-name" is the name of the feed).

Adafruit IO REST Client Library

The Adafruit IO Arduino library (<http://adafru.it/fpd>) is a simple library for sending and receiving the latest value for a feed.

This library uses the send (<http://adafru.it/iRb>) and last (<http://adafru.it/iRb>) Adafruit IO REST API calls and takes care of all the work to use the Adafruit IO REST API. The REST client library supports the following networking platforms/hardware: Adafruit CC3000 (<http://adafru.it/iRa>)

Adafruit FONA (<http://adafru.it/dFz>)

ESP8266 Arduino (<http://adafru.it/eSH>)

Generic Arduino Client Interface (<http://adafru.it/fpb>) (including Ethernet shield and similar network hardware)

To install the library you can use the Arduino library manager (<http://adafru.it/flm>) or download the library from

GitHub (<http://adafru.it/fpd>) and manually install it (<http://adafru.it/dNR>).

Finally make sure you have any required libraries for your network hardware installed.

**CHAPTER
APPLICATION**

CHAPTER
ADVANTAGES AND DISADVANTAGES

CHAPTER

REFERENCES

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